energy_analysis

May 30, 2026

1 Energy usage analysis

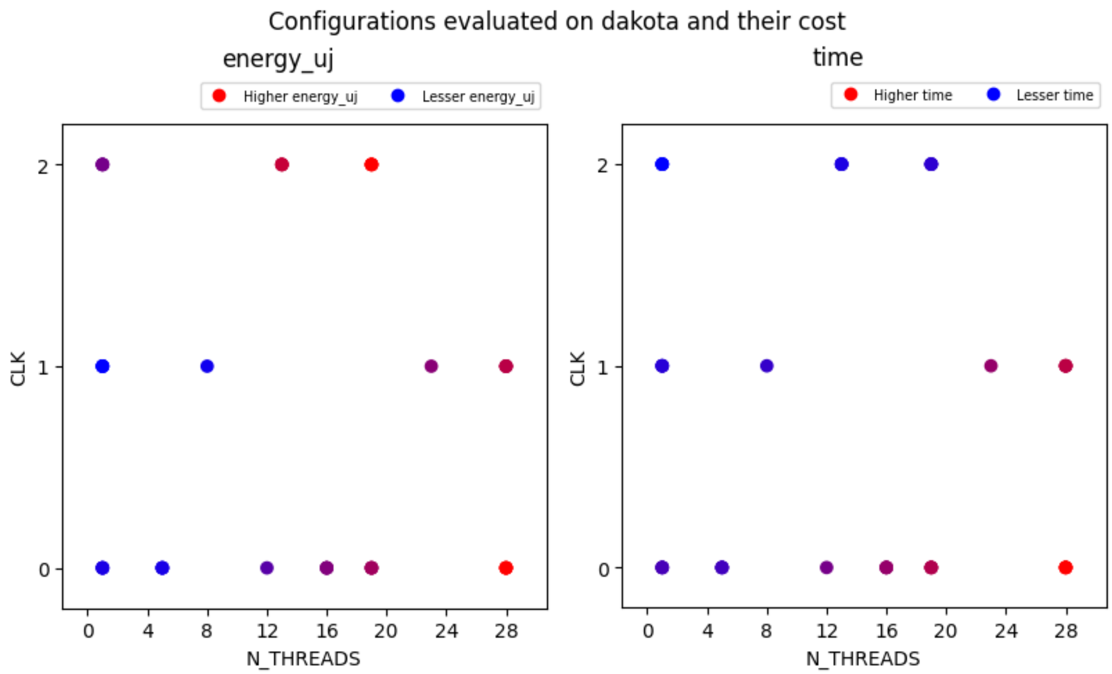
imports

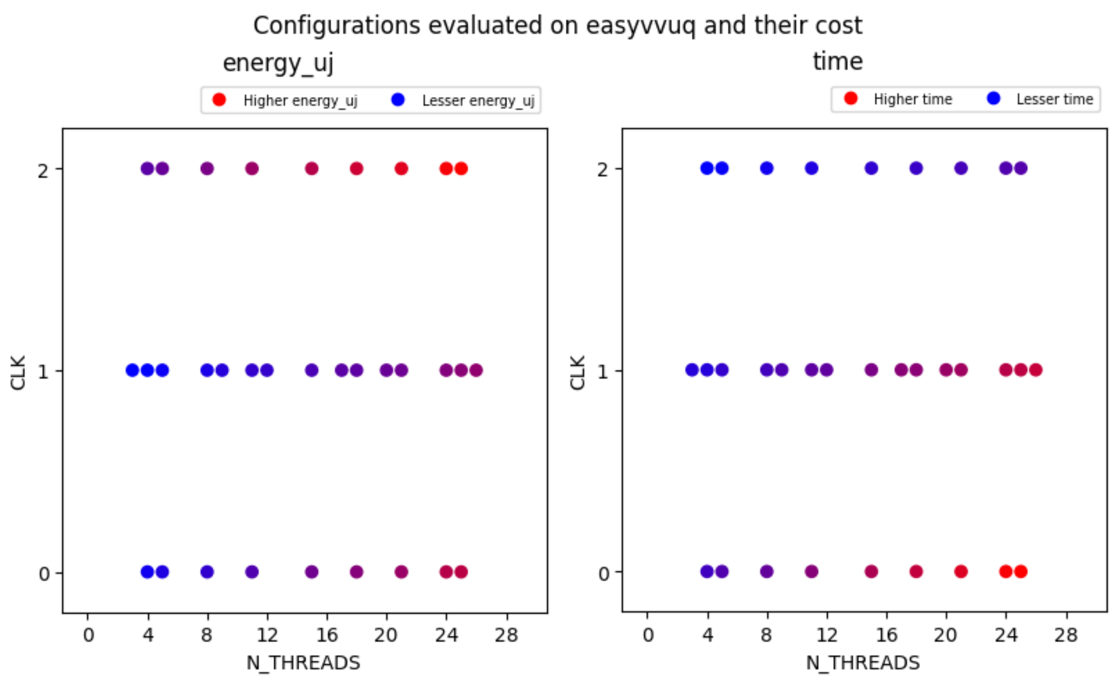
1.0.1 Restore data from most recent runs

```
/home/mmachado/HPC/uq_sbac/.venv/lib/python3.14/site-
packages/cerberus/validator.py:618: UserWarning: These types are defined both
with a method and in the'types_mapping' property of this validator: {'integer'}
  warn(
```

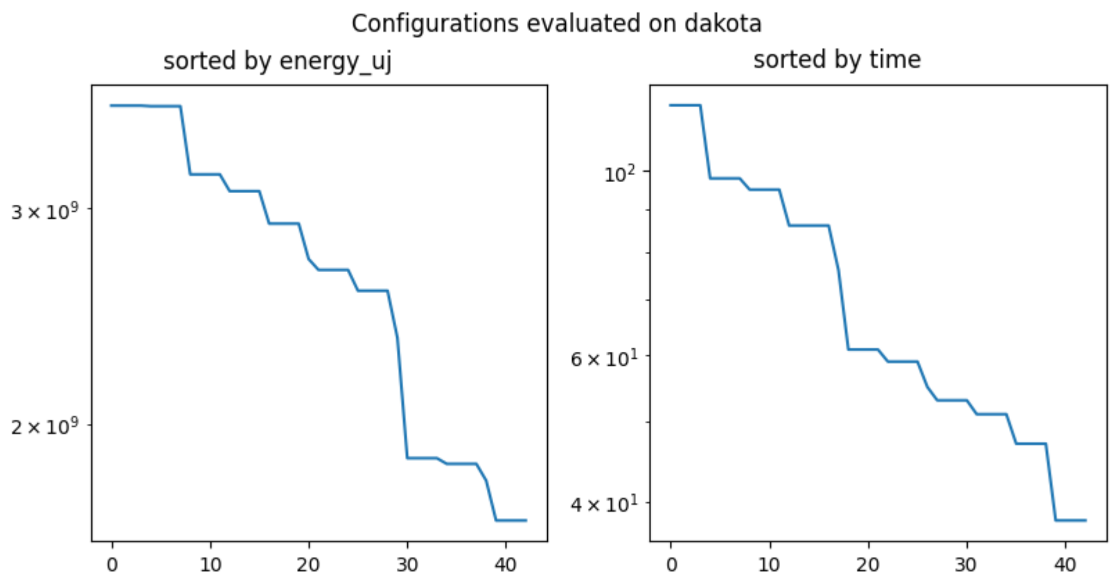
1.1 Plots

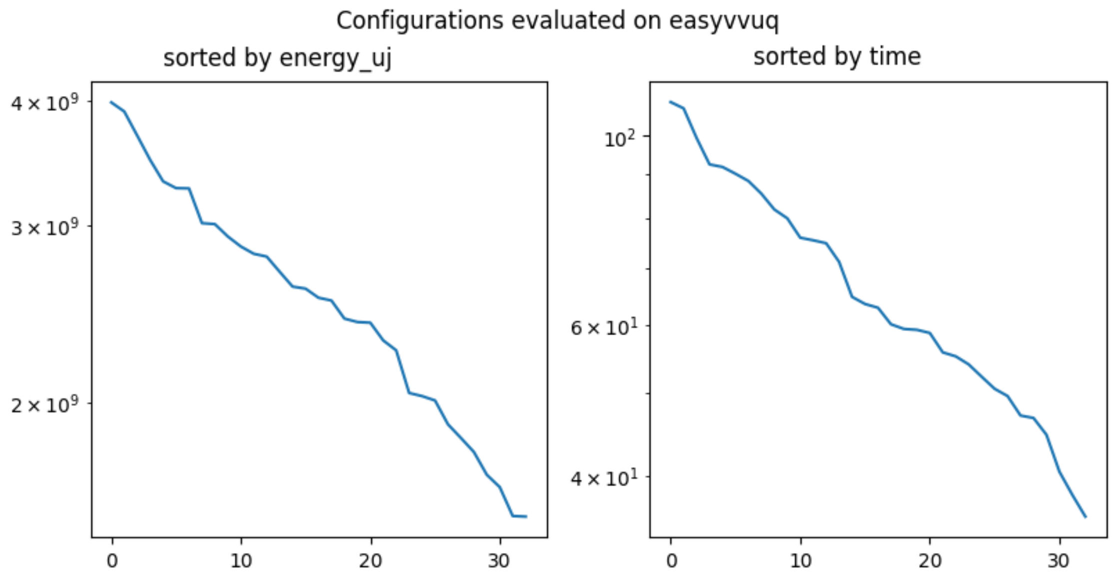
1.1.1 Which configurations consumed less energy or time in total



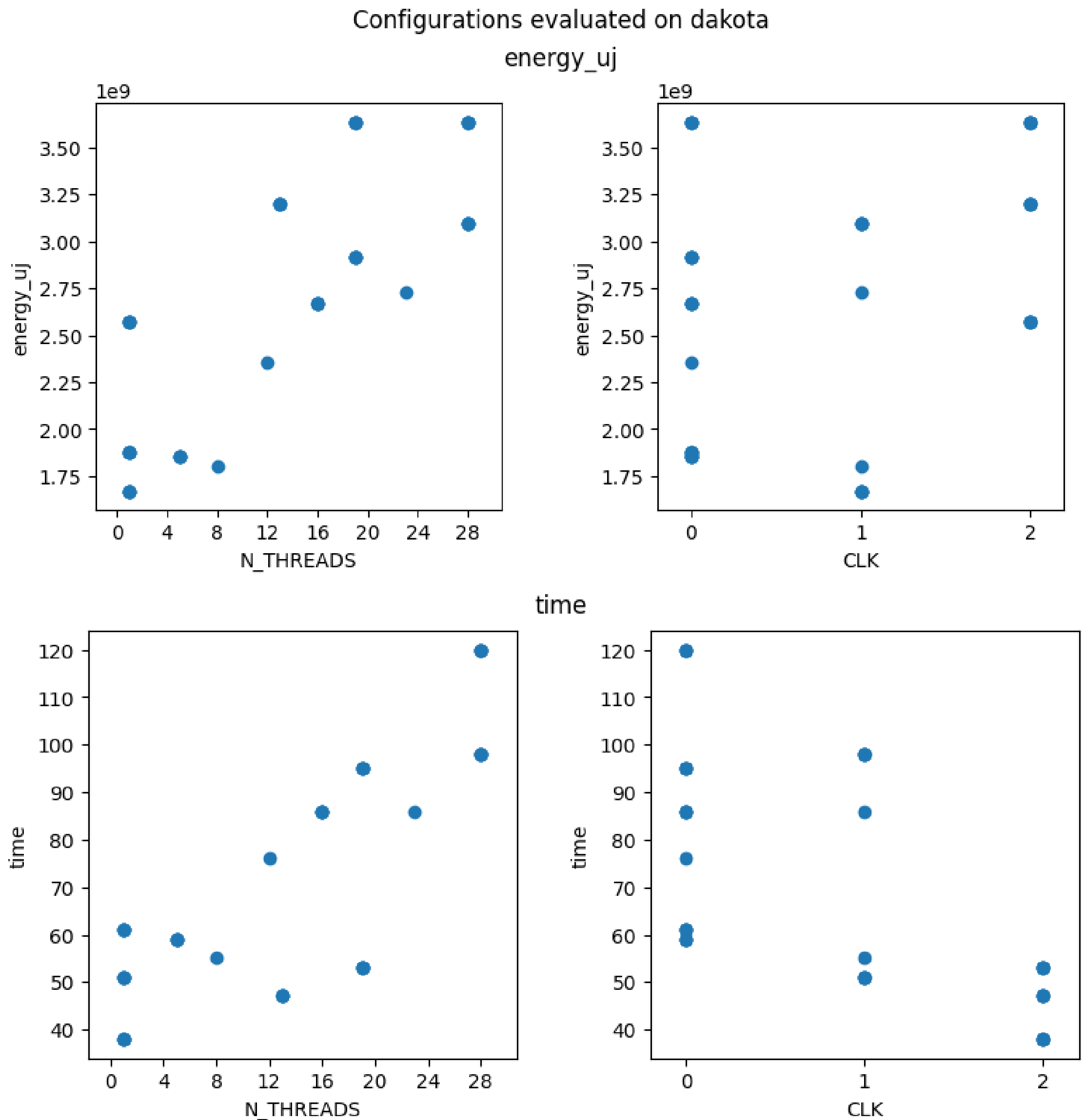


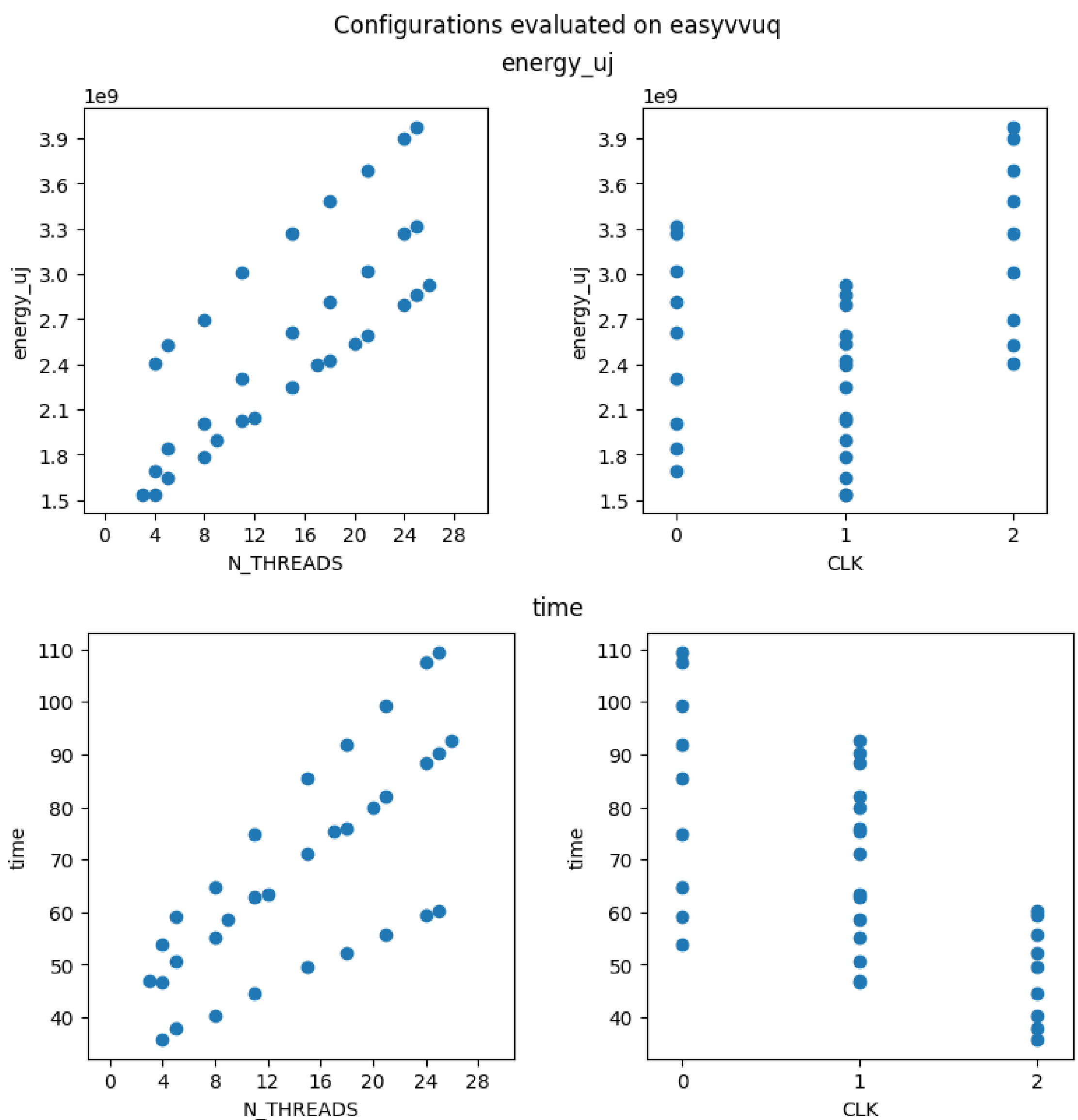
1.1.2 What is the difference between the best and worst configurations?




Configurations evaluated on easyvvuq
sorted by energy_uj
sorted by time
4×10^9
3×10^9
2×10^9
10^2
6×10^1
4×10^1
0
10
20
30

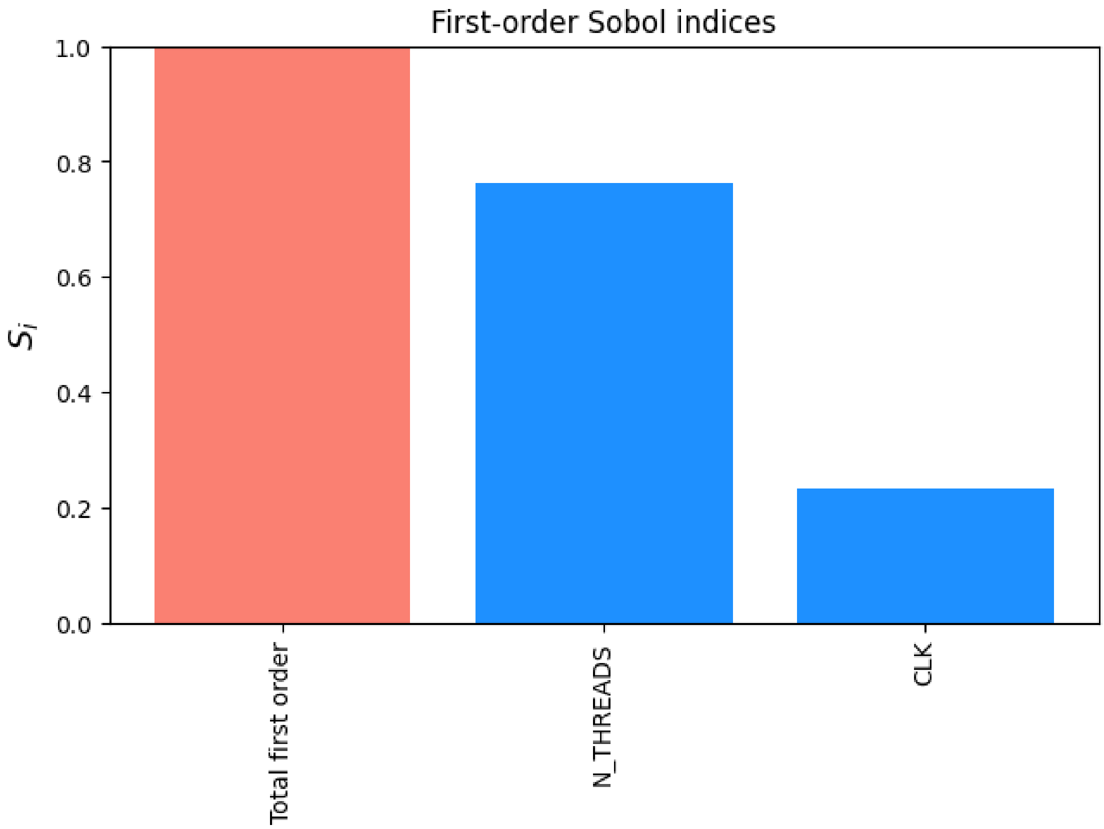
1.1.3 View of the resulting energy usage and time against each parameter

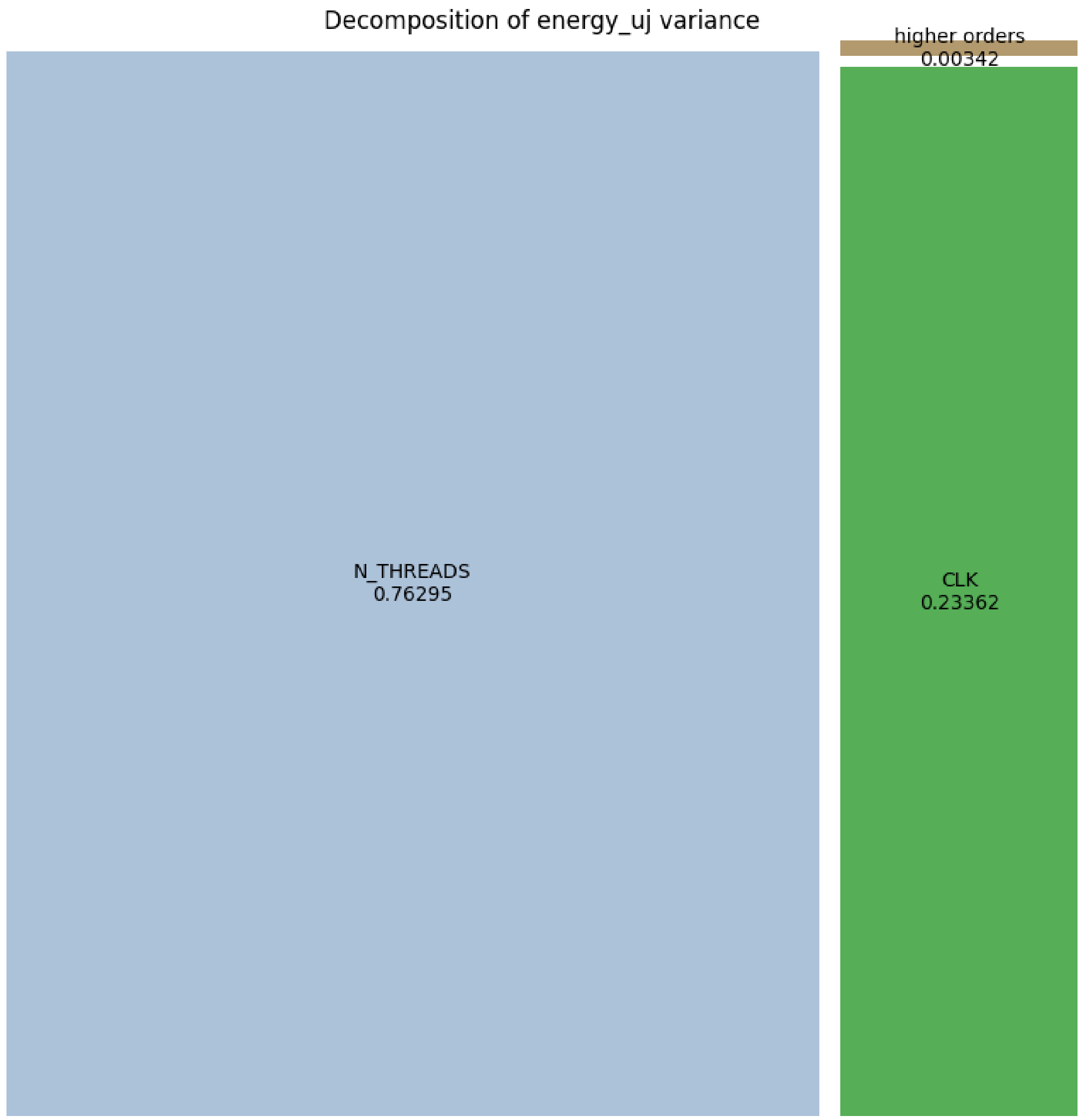



Configurations evaluated on easyvvuq
energy_uj
1e9
3.9
3.6
3.3
3.0
2.7
2.4
2.1
1.8
1.5
energy_uj
0
4
8
12
16
20
24
28
N_THREADS
1e9
energy_uj
0
1
2
CLK
time
110
100
90
80
70
60
50
40
time
0
4
8
12
16
20
24
28
N_THREADS
time
0
1
2
CLK

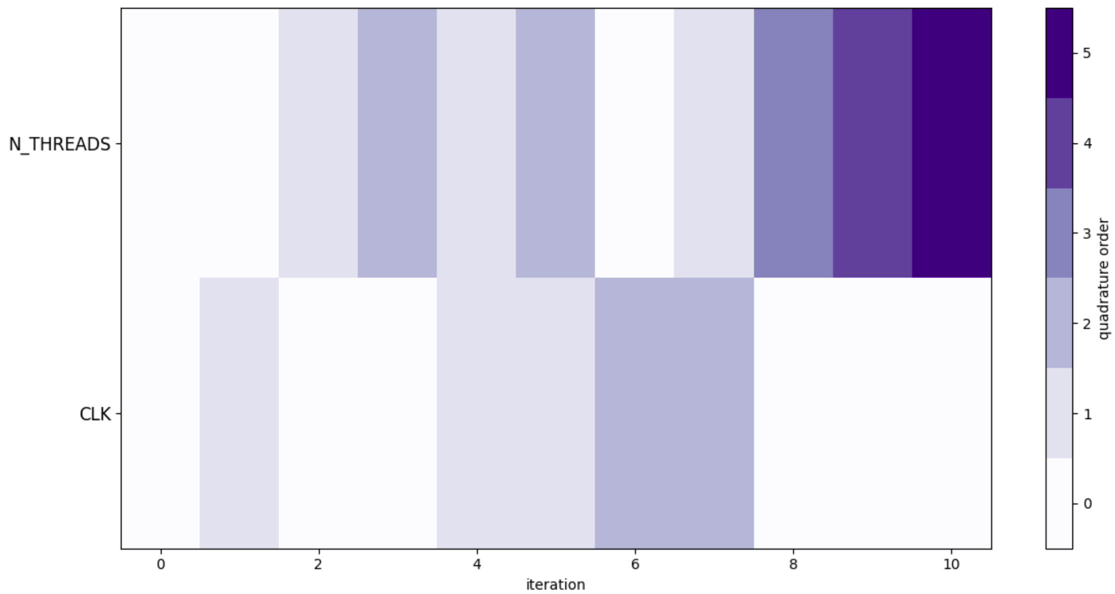
1.2 The following could be generated only from easyvvuq, for now

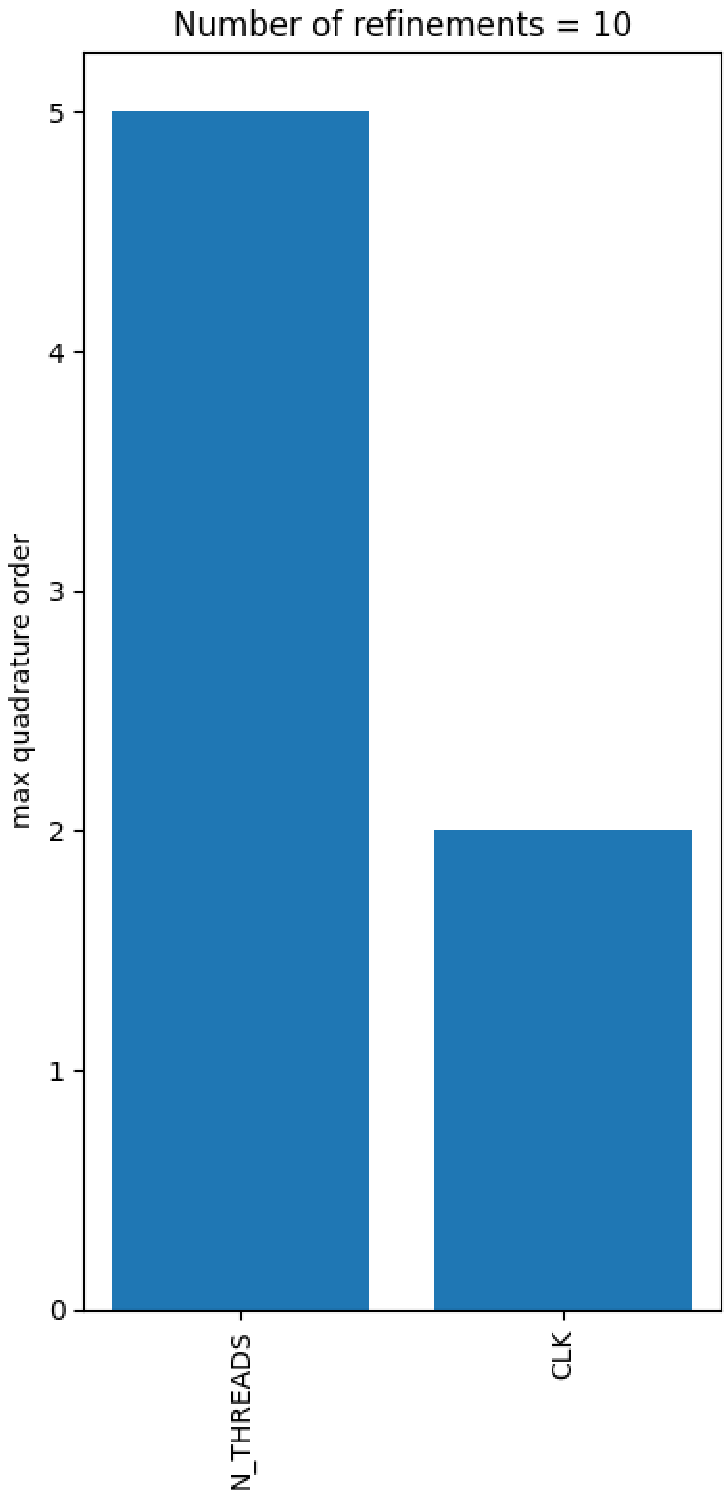
1.2.1 First order sobol indices and decomposition treemap

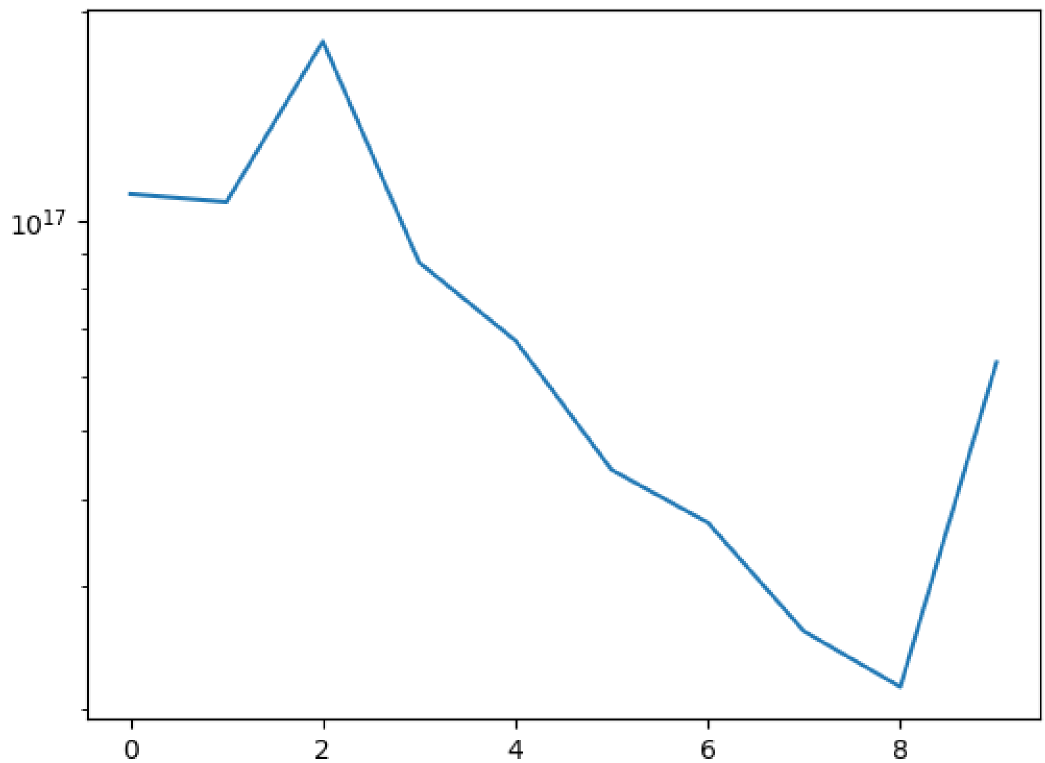



Decomposition of energy_uj variance
higher orders
0.00342
N_THREADS
0.76295
CLK
0.23362

1.2.2 Adaptation table, histogram and errors




Number of refinements = 10
max quadrature order
0
1
2
3
4
5
N_THREADS
CLK



1.2.3 Uncertainty amplification and stat convergence

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Mean CV input = 68.6791 %
Mean CV output = 26.7429 %
Uncertainty amplification factor = 0.2674/0.6868 = 0.3894
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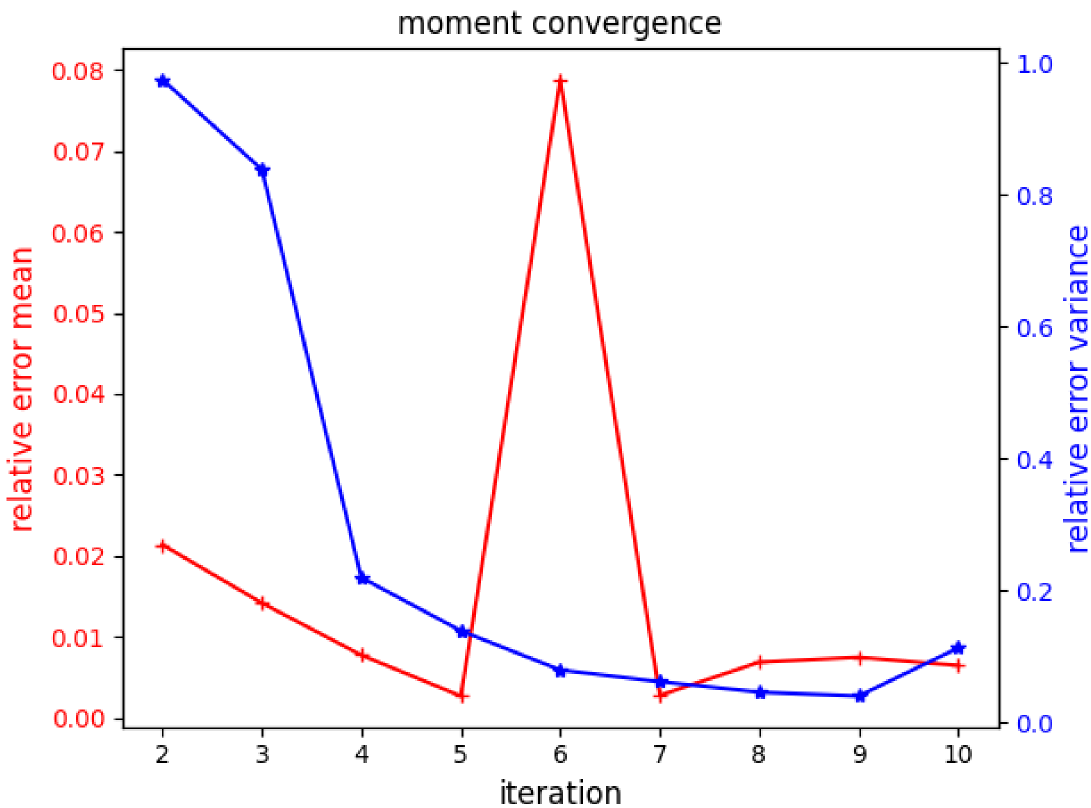
```


moment convergence
relative error mean
relative error variance
iteration
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.0
0.2
0.4
0.6
0.8
1.0
2
3
4
5
6
7
8
9
10